DATASTRAW ASSESSMENT TEST

Hiring Assignment: Build an AI Voice CX Agent for a D2C Brand

Submission deadline: 4 days | Expected effort: Approximately 1.5–2 days of focused work

1. What You'll Build

Build a browser-based AI Voice Customer Support Agent for a fictional D2C skincare brand, Aura Skincare.

When an evaluator opens your application and clicks Start Call, they should be able to speak naturally with the AI agent using their computer's microphone and hear the agent respond through their speakers.

The agent should be able to:

- Have a natural voice conversation with a customer.
- Answer questions using Aura Skincare's brand and policy information.
- Look up live order information when required.
- Follow brand policies and avoid making promises outside those policies.
- Recognise when it does not have enough information to answer something.
- Generate a transcript and structured summary after the call.

> **Note on Telephony Setup**
> You are not required to set up telephony or purchase a phone number. The entire experience can run directly in the browser.

2. Core Requirements

A. Voice Conversation

The customer should be able to speak to the agent and receive a spoken response. The agent should:

- Understand normal conversational speech.
- Respond naturally and concisely (preferably in an Indian voice).
- Maintain context throughout the conversation.
- Avoid unnecessary repetition.
- Have reasonable response speed without long, awkward pauses.

Persona: Aria from Aura Skincare — friendly, professional and concise Indian customer support specialist.

B. Brand Knowledge, Guardrails & Graceful Degradation

The agent must use the information provided below to answer customer questions accurately. It should also follow the brand's policies rather than simply agreeing to whatever the customer asks.

Policy Enforcement Example: Customer: "I bought this 20 days ago and have opened it. Can I return it?"
The agent should politely explain that the request falls outside the return policy rather than promising a refund.

Out-of-Scope Example: Customer: "Can you book me a flight to Goa?"
The agent should politely explain that it can only help with Aura Skincare-related queries.

Graceful Degradation & Edge Cases: If the customer mumbles, audio is unclear, or an order ID does not exist in the database, the agent should handle it gracefully (e.g., "I couldn't locate an order with that number, could you please repeat or verify the ID?") rather than hallucinating or crashing.

C. Order Lookup — Tool / Function Calling

The agent must be able to retrieve order information from the provided mock order database. At minimum, implement get_order_details(order_id).

For example: Customer: "Can you tell me where my order ORD-101 is?"
The agent should identify that it needs order information, call the relevant tool, retrieve the order details and communicate the result naturally to the customer.

The agent should also handle: invalid order IDs, missing order IDs, and orders that are not eligible for a requested action. You may implement additional tools if you think they improve the experience, but this is not required.

D. Post-Call Summary

When the call ends, the application should display:

- **1. Call transcript:** A chronological transcript showing both the customer and agent.
- **2. Structured call outcome:** Formatted in JSON (the exact structure is up to you, as long as key data is captured):

```
{
  "customer_intent": "ORDER_TRACKING",
  "order_id": "ORD-101",
  "resolution_status": "RESOLVED",
  "call_summary": "Customer asked about the delivery status of ORD-101. The order is out for delivery and
expected by 6 PM today."
}
```

E. Simple Testing Interface

The evaluator should be able to easily test your agent. At minimum, include:

- Start Call button.
- End Call button.
- **Live State Indicator:** Clear visual feedback of the agent's current state (Listening / Thinking / Speaking).
- **Test Orders Helper:** A visible panel or helper card directly on the page showing the available sample order IDs (ORD-101, ORD-102, ORD-103) and their details, so evaluators can test order lookups immediately without having to reference external documents.
- **Call Log & Summary:** Transcript or call log after the conversation concludes.

The interface does not need to be highly designed. Functionality and user experience matter more than visual complexity.

3. Aura Skincare — Brand Information

Brand Overview: Aura Skincare is a premium organic Indian skincare brand focused on simple, effective skincare products made with thoughtfully selected ingredients.

Shipping Policy: Free delivery on orders above ₹499. Orders below ₹499 have a ₹50 shipping fee. Standard delivery takes 3–5 business days.

Return & Refund Policy: Returns are accepted within 7 days of delivery for unopened, unused products in original packaging. Damaged or defective products must be reported within 48 hours of delivery with photos for replacement.

Cancellation Policy: Orders can be cancelled only while their status is Processing. Once an order is Shipped or Out for Delivery, it cannot be cancelled (customers may refuse delivery at doorstep).

Cash on Delivery: COD is available for orders up to ₹2,500. Customers can pay by cash or UPI at the doorstep.

4. Mock Order Database

Order ID	Customer	Product	Value	Status	Notes
ORD-101	Priya Sharma	Vitamin C Serum (30ml)	₹699	**Out for Delivery**	BlueDart — BD-982103. Expected by 6 PM today
ORD-102	Rahul Verma	Hydrating Sunscreen SPF 50	₹499	**Delivered**	Delhivery — DL-441029. Delivered 14 days ago
ORD-103	Ananya Patel	Green Tea Face Wash + Toner	₹850	**Processing**	Ordered 3 hours ago. Eligible for cancellation

5. Technology

The technology stack and architecture are completely up to you. You may use any tools, frameworks, APIs, or platforms that you are comfortable with.

Common architectural approaches you can take:

- **Modular Pipeline:** Composing custom Speech-to-Text (STT) → LLM reasoning & tool execution → Text-to-Speech (TTS) audio streaming.
- **Realtime Multimodal APIs:** Direct speech-to-speech / audio streaming models.
- **Voice AI Frameworks:** Any existing open-source or managed voice frameworks.

Languages & Stacks: You may use React, Next.js, Vue, Python, Node.js or any other technologies you prefer.

We are not evaluating you on whether you use a particular vendor. We are more interested in how you approach the problem, the natural feel and latency of the resulting experience, and your understanding of what you built.

6. AI Tools Are Encouraged

You are absolutely encouraged to use AI-assisted development tools. You may use tools such as:

- Cursor, Claude Code, ChatGPT / Codex, GitHub Copilot, Windsurf, Google Gemini, or other AI coding/research tools.

This is how modern engineering teams increasingly work. However, you should understand the system you submit. During the evaluation, we may ask you to explain:

- Why you chose your architecture.
- How the voice pipeline works.
- How your agent decides when to use a tool.
- How you implemented the brand guardrails.
- What you would change or improve.

We are not looking for code written entirely from scratch. We are looking for someone who can use modern tools effectively and understands what they are building.

7. Deployment

Your application should be publicly accessible so that we can test it without setting up your development environment.

- Vercel, Render, Railway, Fly.io, Netlify, Hugging Face Spaces, or any equivalent platform.

You do not need to purchase a domain. If your chosen AI/voice provider requires API credits, free trials or free tiers are completely acceptable.

8. Optional — Go Beyond the Basics

The following are not required, but you can implement them if you think they meaningfully improve the product:

- Barge-in / interruption handling — agent stops speaking when the customer interrupts.
- Hinglish support.
- Multiple tools.
- Better handling of ambiguous customer requests.
- Improved call experience or UI.
- Multi-brand support.
- Very low latency.
- Anything else you think would make this a better real-world customer support agent.

Don't add features just for the sake of adding them. We value thoughtful decisions more than feature count.

9. Tell Us How You Think

Along with your application, include a short section in your README answering the following:

- **1.** Why did you choose your particular architecture and technology stack?
- **2.** What was the most difficult part of the assignment, and how did you solve it?
- **3.** If you had one more week to work on this, what would you improve first and why?
- **4.** Imagine this agent is handling 1,000 customer conversations a day. What do you think would need to change or improve?

There are no 'correct' answers here. We want to understand how you think about the product and the engineering behind it.

10. What to Submit

Please submit:

- **1. Public Application URL:** A working link where we can open the application, allow microphone access and test the voice agent.
- **2. GitHub Repository:** Source code, README.md (including your answers to Section 9), architecture/approach, setup instructions, and .env.example.
- **3. 3–5 Minute Demo Video:** Show the voice agent working, at least one order lookup, one policy-related conversation, your post-call transcript and summary, and a brief architecture walkthrough.
- **4. Short Approach Note:** Include 2–3 sentences summarising your approach.
- **5. LinkedIn Profile:** Please include a link to your LinkedIn profile and make sure the link is accessible.

11. Submission

Please send your submission to:

> **WHERE TO SEND YOUR SUBMISSION**
> Please submit your completed assignment to:
> • Primary: ozair.shaikh@datastraw.in | aryan.jaiswal@datastraw.in
> • CC: talent@datastraw.in
> • Subject: AI Voice Agent Assignment - [Your Full Name]
>
> Please include in your email:
> 1. Application URL
> 2. GitHub Repository
> 3. Demo Video Link
> 4. LinkedIn Profile Link (ensure working)
> 5. 2–3 sentence approach note

What We're Looking For

We're not expecting a production-ready voice AI platform. We're looking for someone who can take an open-ended problem, learn what they need to learn, use modern AI tools effectively, build a working solution, and think about both the technology and the customer experience.

Good luck — and have fun building!